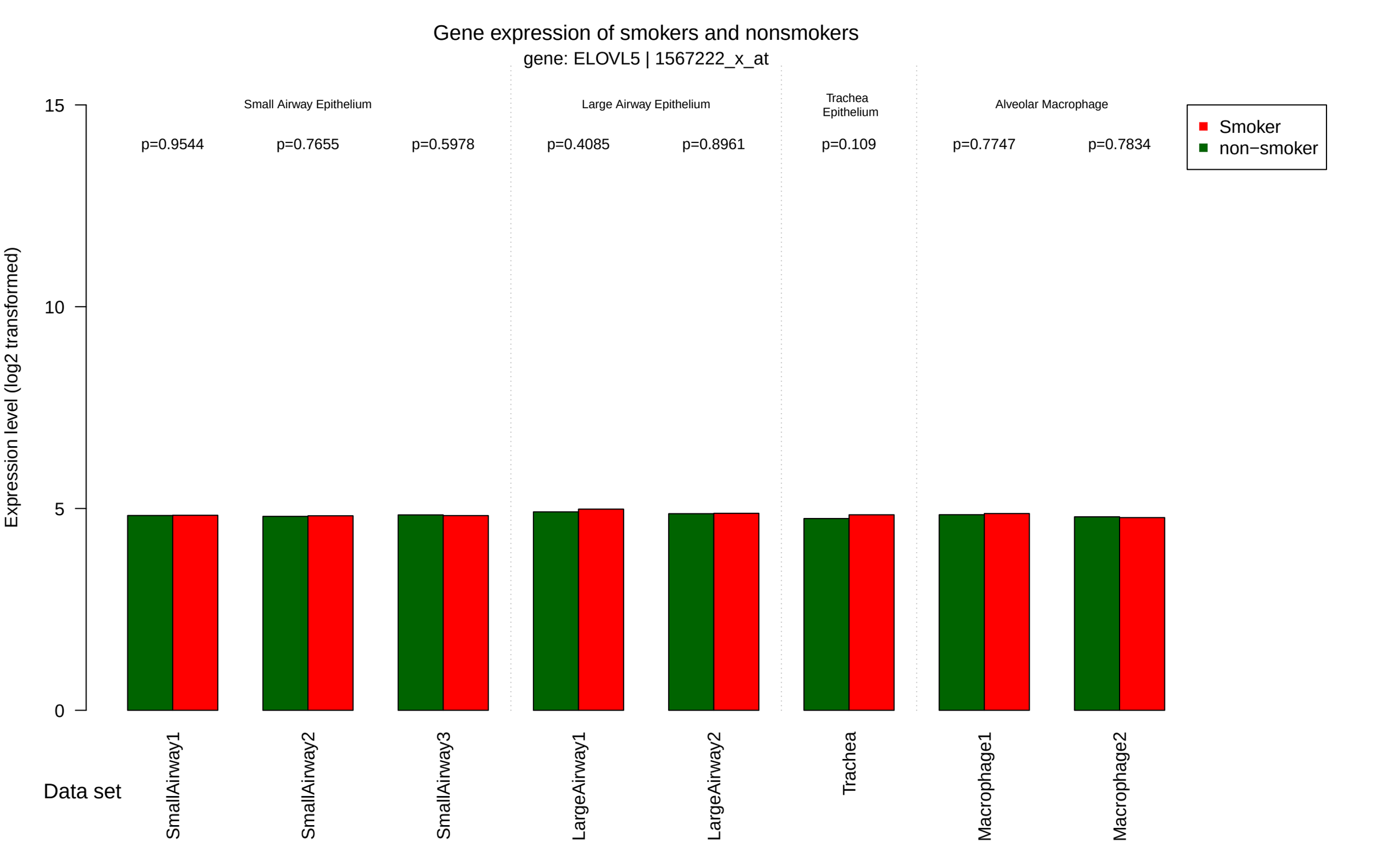
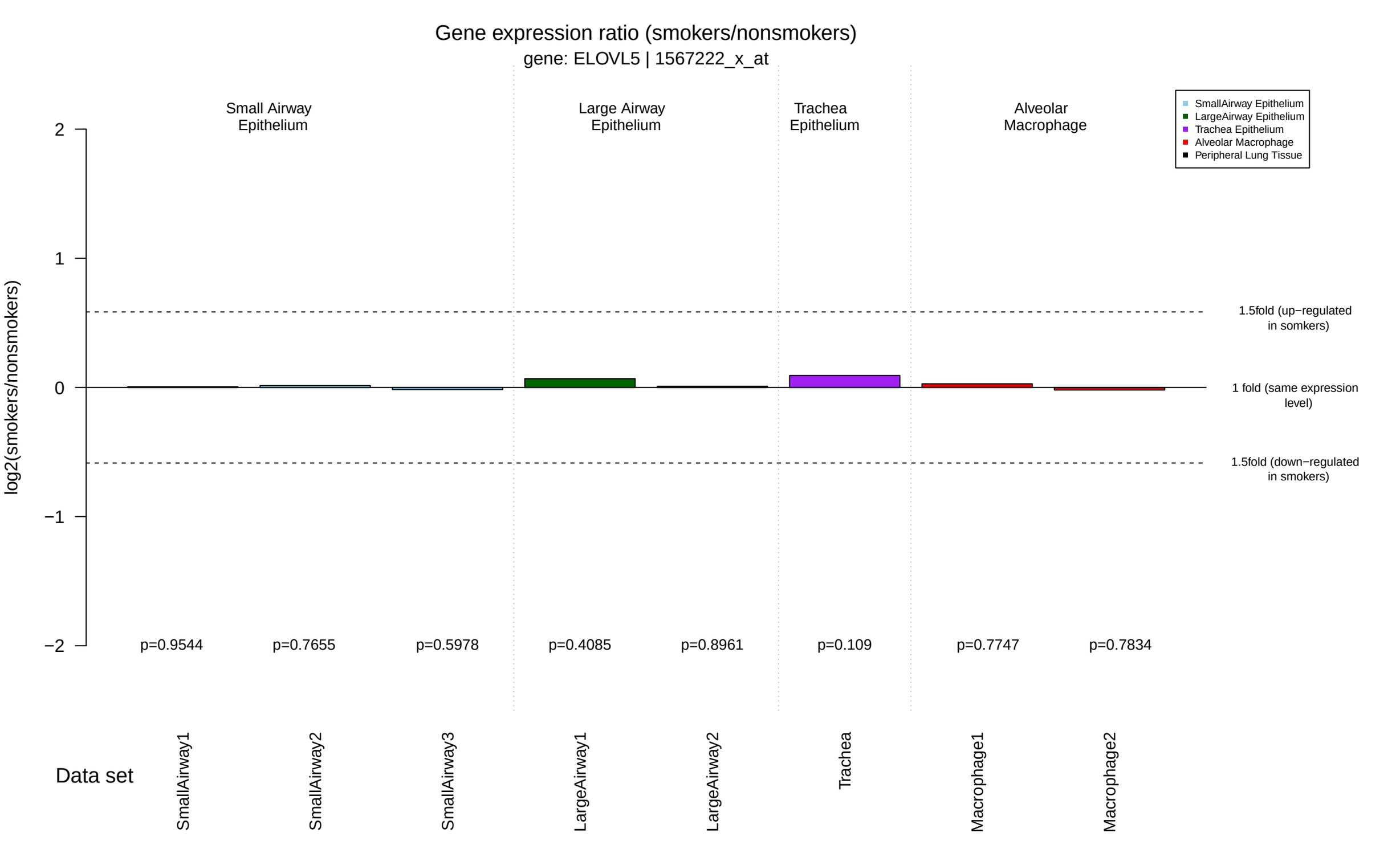
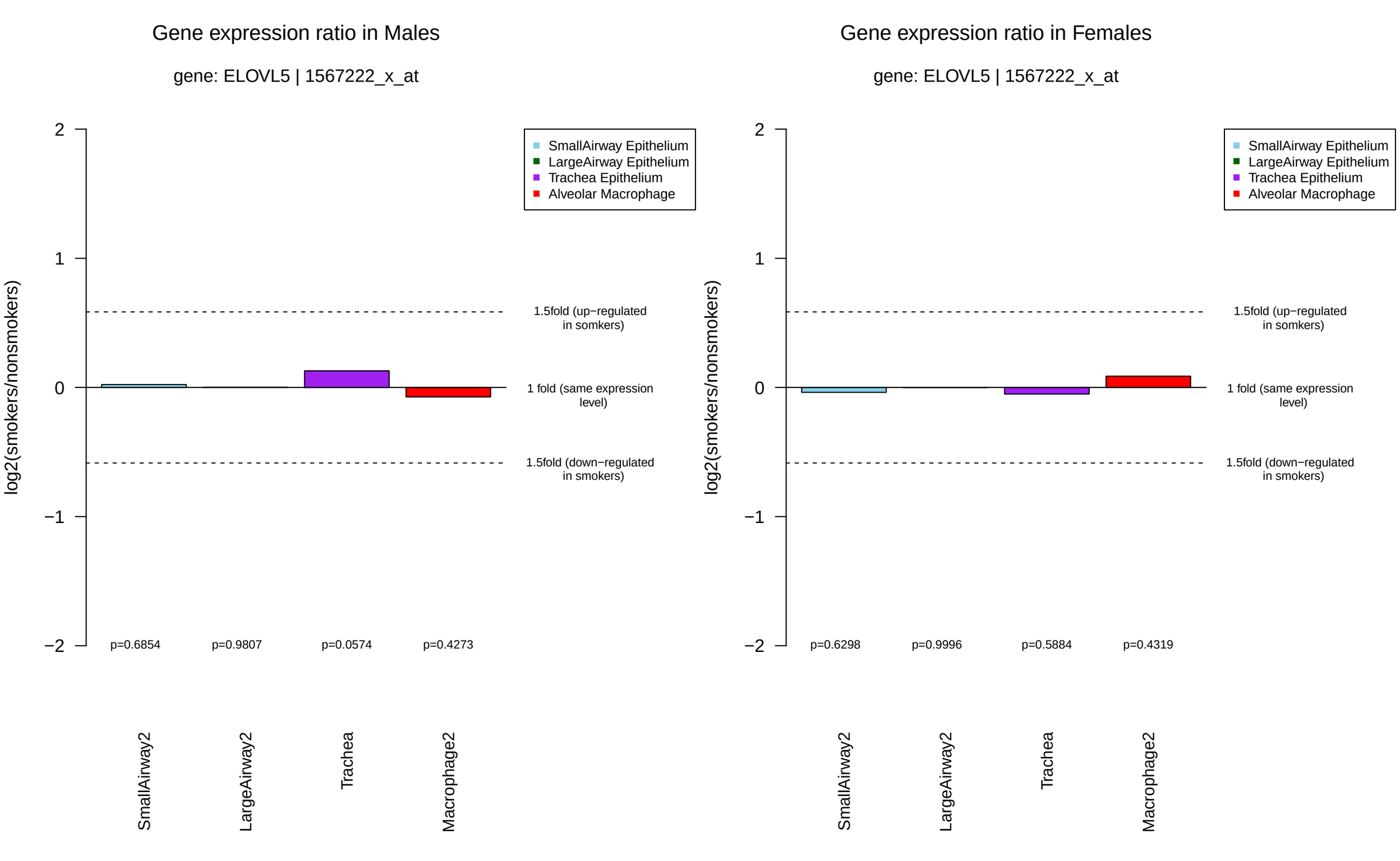
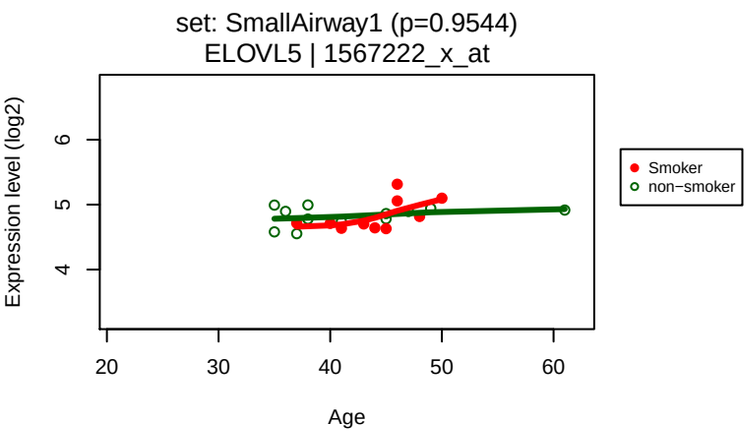
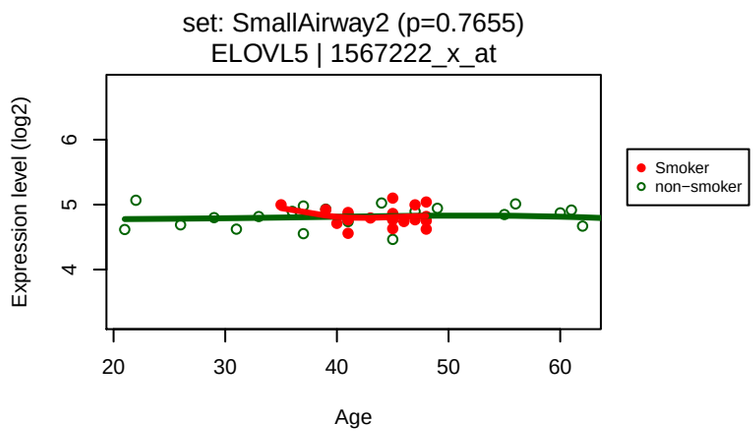
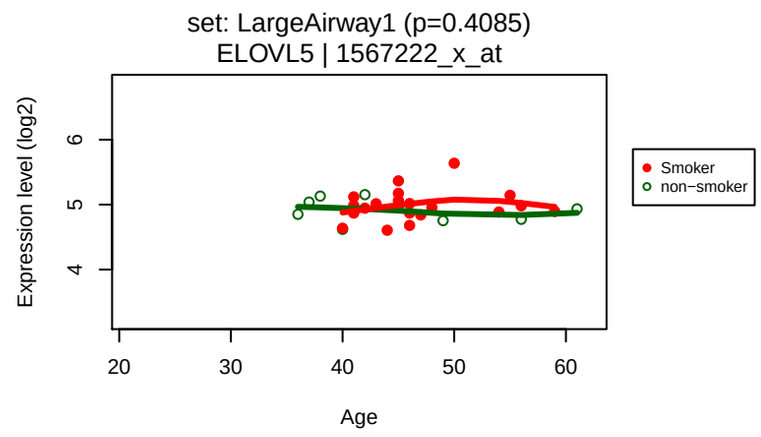

Gene expression of smokers and nonsmokers
gene: ELOVL5 | 1567222_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9544
p=0.7655
p=0.5978
p=0.4085
p=0.8961
p=0.109
p=0.7747
p=0.7834
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

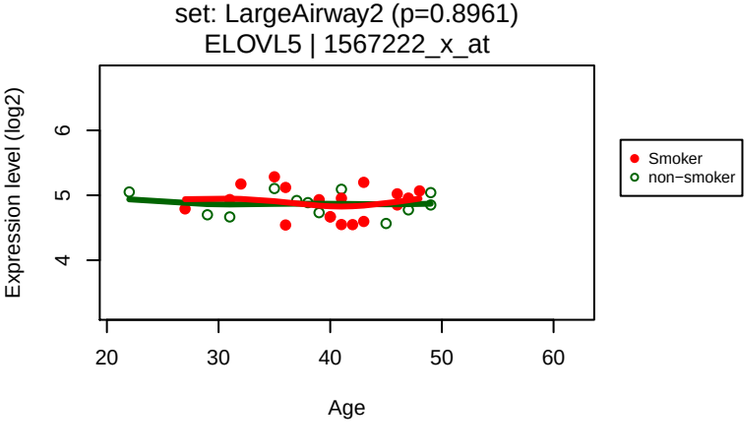

Gene expression ratio (smokers/nonsmokers)
gene: ELOVL5 | 1567222_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9544
p=0.7655
p=0.5978
p=0.4085
p=0.8961
p=0.109
p=0.7747
p=0.7834
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

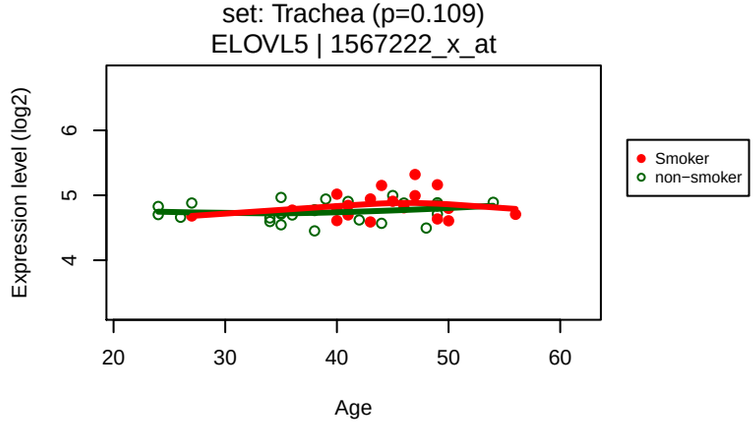

Gene expression ratio in Males
gene: ELOVL5 | 1567222_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6854
p=0.9807
p=0.0574
p=0.4273
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ELOVL5 | 1567222_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6298
p=0.9996
p=0.5884
p=0.4319
SmallAirway2
LargeAirway2
Trachea
Macrophage2

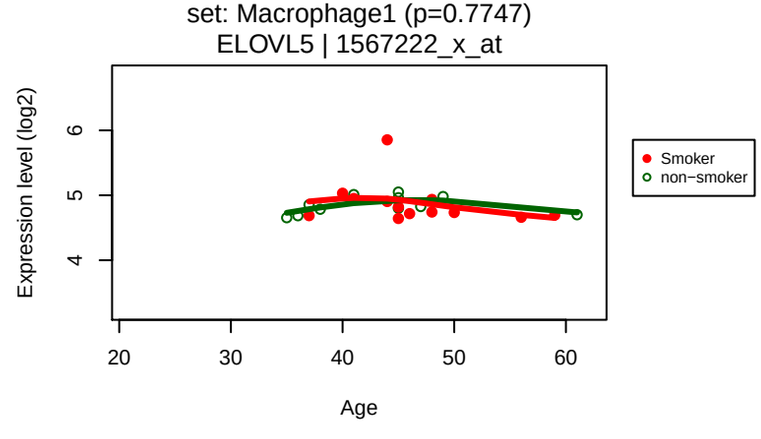

set: SmallAirway1 (p=0.9544)
ELOVL5 | 1567222_x_at
Expression level (log2)
Age
Smoker
non−smoker

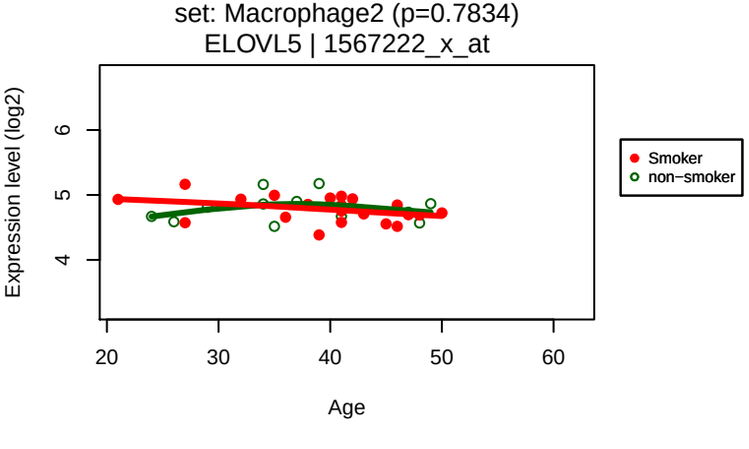

set: SmallAirway2 (p=0.7655)
ELOVL5 | 1567222_x_at
Expression level (log2)
Age
Smoker
non−smoker

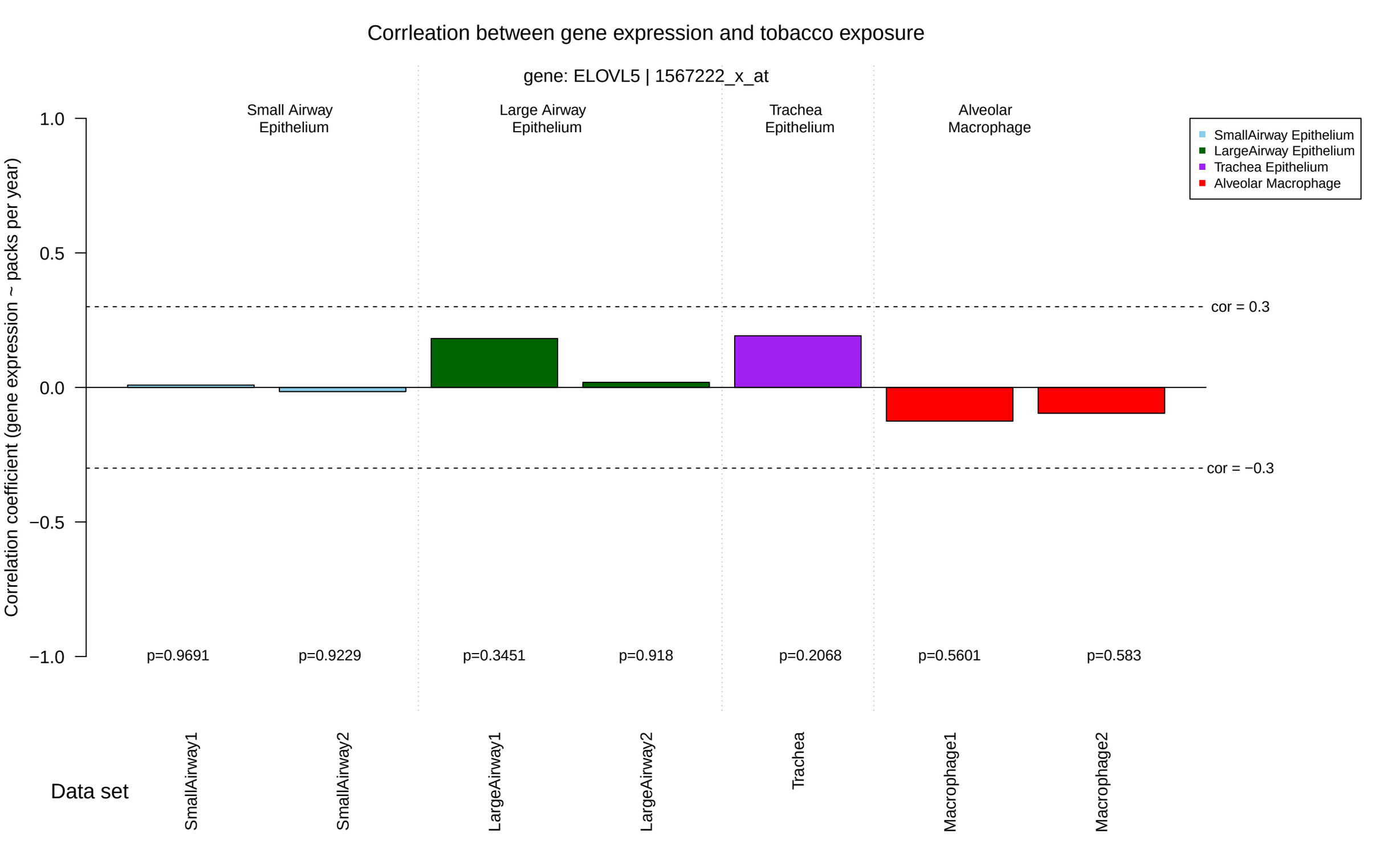

set: LargeAirway1 (p=0.4085)
ELOVL5 | 1567222_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8961)
ELOVL5 | 1567222_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.109)
ELOVL5 | 1567222_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7747)
ELOVL5 | 1567222_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7834)
ELOVL5 | 1567222_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ELOVL5 | 1567222_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9691
p=0.9229
p=0.3451
p=0.918
p=0.2068
p=0.5601
p=0.583
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2